

Similarity methods — the master table

One row per instrument. Every number is measured and appears in the Final Project Report (68 pp) or an executed notebook; figure names refer to project files and the corrections pack.

method / term	what it asks, in plain words	the math	where it was run / figure	obstacles and tuning on the way	result — and what it proved	implication / engineering outcome
Raw cross-space cosine	Do two encoders give the same item the same COORDINATES?	$\cos(xA_i, xB_i)$ on the raw vectors, no fitting	Exp B pair; every pair · report p34	none — this is the honest baseline	≈ 0.00 everywhere. Coordinates are NEVER shared; raw agreement is exactly chance	kills naive interop: you cannot mix vectors from two models directly — something must translate
Ridge linear map (W)	Can ONE matrix translate space A into space B?	$W = \text{argmin} \ XW - Y\ _F^2 + \alpha \cdot \ W\ _F^2$, closed form; $\alpha = 1e-2$	Exp B (MobileCLIP→SigLIP); E-series (DINOv2→bge); all hub entry maps	α swept on validation; rows/dim ≥ 5 pre-registered floor (9.1 achieved); 'α cannot substitute for width' later corrected — it can (E.8/E.9)	B: R ² 0.592, cosine to twin 0.900. The relation between worlds is LINEAR to first order	the entire deployable is one matrix: 0.79 MB Core ML fp16, no GPU, no retraining
Retrieval R@K vs ceiling	Does the translated vector still FIND the right item among impostors?	rank $\cos(q, t_j)$ over the gallery; report R@1 as % of a jointly-trained ceiling; chance = 1/N	B: 1,000-image gallery; E-series ladder; hub demo (G2)	ceiling must be re-measured on the SAME rows with the SAME caption treatment — mismatched, it read 55.2%; matched, 40.9%	B: 88.4% of ceiling. Ladder: 40.9/53.5/58.3% (+15.3 pts per decade of encoder capacity, R ² 0.949)	%-of-ceiling became the project's common currency — every transfer figure is a level against an honest reference
Verification AUC	Same item or different — can the mapped vector tell?	ROC over $\cos(\text{mapped}, \text{candidate})$; same-vs-impostor pairs	B: AUC 0.999 (vs 0.848 on hardest-of-1,000); hub V1: 0.9994	0.999 alone would scream overfitting — defused by showing the SAME map scores far lower on harder questions, and the no-adapter control at 0.477	near-perfect same/different separation with one matrix; control collapses to chance	cheap identity check usable as a runtime guard on any cross-encoder pipeline
Distance-rank correlation (shape agreement)	Is near still near and far still far — IGNORING coordinates?	Spearman rho (and Pearson) between the two spaces' pairwise cosine-distance lists; nothing fitted	report p34 (B pair: +0.681 / +0.763); Fig A15b — all 21 pairs of the seven-space hub	invariant to rotation, scale, even dimension — so no alignment step to get wrong	shape is shared where coordinates are not: cross-modal ≈ +0.29, siblings higher, bge–SBERT tops at +0.764	proves the map EXPOSES structure rather than creating it — the caption of every diagram since
k-NN overlap, multi-k	How many of an item's nearest neighbours are the SAME items in both spaces — and at what scale?	$ \text{NN}_k A(i) \cap \text{NN}_k B(i) / k$, vs chance k/N ; swept $k = 1/10/100/500$	Table C12h; Fig 1 of the corrections pack (log-scale decay)	chance must be k/N of the kNN set (2.5%), not the retrieval gallery's 0.001 — an easy and consequential mix-up	67.1% at $k=10$ vs 2.5% chance (27×); every pair decays toward chance by $k=500$	'local-over-global' quantified — the finding that names what 'same world' actually means
CKNNA	Mutual-nearest-neighbour alignment as k grows (the PRH paper's own metric)	mutual-kNN agreement, normalised; curve over k	C.13.5 / C.13.9 — all 21 pairs	one metric could be a fluke — so it is one of FOUR independent instruments pointed at the same question	21 of 21 pairs score higher locally than at the global limit (e.g. 0.917 at $k=10 \rightarrow 0.169$)	unanimity across pairs turns a tendency into a law of the corpus
CKA (+ hubness correction)	Do two networks' LAYERS carry similar information?	centred kernel alignment on layer activations; row/column marginals divided out	Exp A2: GPT-2 vs Pythia-160M, 10,000 samples; layer panel p23	the panel looked blocky — a rank-one hubness component carried 97% of matrix energy; dividing marginals out lifted diagonal enrichment to 1.16 vs 0.98	trained-vs-trained 0.359 vs trained-vs-random 0.067 — a 4–5× learning effect across architectures	the first evidence in-project that convergence exists at all — and a worked lesson in correcting a metric before believing it
Procrustes (orthogonal / similarity)	Is a pure ROTATION (+scale) enough to align two spaces?	$\min \ XR - Y\ _F$ over orthogonal R ; similarity adds one scale	K1 — raw spaces, no PCA anywhere	an earlier run compared mismatched scalings (G6); K1 re-ran clean on raw spaces	similarity transform recovers 0.281 vs ridge 0.667 (42%) — rotation is NOT enough; bge–SBERT the near-isometric exception (0.758)	'related, not rigid': the sharedness is linear, not merely rotational — sets the algebra any interop layer needs
CCA shared directions	How many independent DIRECTIONS do two spaces share?	canonical correlation; count directions above threshold of 64	Table C12k	reduced to a common PCA dimension first — documented, since it equalises per-axis variance	cross-modal pairs share 6–32 of 64; siblings and same-objective text pairs share nearly all	a budget for how wide a shared interface can honestly be, per pair
Gromov–Wasserstein (pair-free)	Could the correspondence be found with NO paired examples at all?	optimal transport matching distance structure to distance structure	C.12 — MobileCLIP vs SigLIP	anisotropy actively hurts the SEARCH; scope bounded: two contrastive encoders, weakest-independence pair	39.7% recovery with zero pairs — far above chance	supervision-free alignment is possible in principle; also the second isotropy instance (helps reading, hurts finding)
Whitening (isotropy transform)	If a space is collapsed, is the information gone — or hidden?	$x \rightarrow S^{(-1/2)}(x - \mu)$: divide each direction by its standard deviation	C.11 — the GPT-2 rescue	must be fit on train rows only; the temptation is to call the space 'dead' and skip it	R@1 0.130 → 0.478 = 102.6% of an encoder trained on 1.2B pairs. The geometry hid the signal; it did not destroy it	a free, closed-form preprocessing that turns an unusable embedding into a working one
Effective rank & spectrum	How many directions actually carry variance — and where does a FIT break?	entropy rank $\exp(H(s^2))$ vs ALGEBRAIC rank; singular-value inspection	GPT-2 6.6, BERT 65.0; E.8 / G13–G16 for the fit boundary	the project's own recorded error: entropy rank (573.7) was compared against a prediction about algebraic rank (768) — different quantities, false falsification (G13)	the width cliff sits exactly at ALGEBRAIC rank: sigma drops 10.79 → 0.0054 at $k=768$ and least squares divides by it — a step, not a decline	the rule every hub inherits: $\text{hub_width} < \text{dim}(\text{source})$; ridge or truncation lifts it entirely

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The Hub (whitened-PCA pivot)	Can N encoders share ONE space instead of N ² connectors?	z-score, concat, PCA(512), whiten = H; entry maps $X_k W_k \approx H$; one head $H \rightarrow bge$	G-series; 4/5/7-space configurations; flow_g + demo	width cliff (above); G10: hub text-composition moves transfer 11.2 pt and the target-in-hub overlap is worth +3.0 pt — measured and DECLARED	a head trained on ONE encoder transfers at 92.9–96.7% to encoders it never saw (SigLIP 94.2, ConvNeXt 96.7 — above the family band); 42/42 pairs at 103.6% of bespoke	adding a model = one closed-form solve; nothing else moves. Extensibility as an engineering property, demonstrated live
MLP control	Would nonlinear capacity beat the matrix?	small MLP, trained; vs the closed-form linear map	two independent runs	data-starvation objection retired: retrained at 74 rows/dim with teacher targets — plateau by epoch 10, same answer	0.9183 vs 0.9183, gain +0.0000, twice. Linearity is sufficient under this test	licenses shipping the simplest possible artifact — and keeps the whole system trainable-free
Controls: random map & shuffle	Would ANY map score this well?	random W in the same slot; shuffled pairings	every transfer table carries one	the discipline: a number without its control is not reported	random map → chance (0.001–0.004); honest fits beat shuffled by 0.545–0.863 vs a 0.2 threshold	the reason 96.7% means something — and the first thing quoted when challenged
Cluster bootstrap null (SESOI)	Is 'no effect' a demonstrated null or just low power?	encoder-level resampling; 95% CI + minimum detectable effect vs pre-registered SESOI	K3 — the hub's cost on local structure	encoders resampled, not pairs — six pairs come from four encoders and are not independent	CI −0.038...+0.024, MDE 0.034 against SESOI 0.050 — a bounded, demonstrated null	the difference between 'we found nothing' and 'there is nothing this big to find'
Stratified local-structure audit	Does the hub damage any single space's neighbourhoods?	per-space neighbourhood preservation, degenerate spaces stratified BEFORE averaging	G7 + K2 re-run	first run's verdict was contaminated by pooling GPT-2's degenerate space into the mean — stratify, then average	healthy spaces neutral (SBERT 0.135); GPT-2 0.262 vs published 0.257 — reproduces; Spearman +0.891	'stratify before averaging' became a standing project rule

Achievements, and what actually ships

Scientific: convergence demonstrated across three supervision regimes and four objectives, with the sharedness located precisely — LOCAL structure over global, objective over lineage over modality (+53× / +44× / +29×), isotropy helping reading and hurting finding. Nine findings; eleven explanations falsified on pre-registered criteria, two of them the project's own, kept on the record. **Engineering, final implementations:** (1) the 0.79 MB Core ML adapter — one fp16 matrix, deployable on-device, 88.4% of a jointly-trained ceiling; (2) the 512-d hub — seven entry maps and one head, rebuildable in seconds on CPU, extensible by one closed-form solve per new encoder, demonstrated with ConvNeXt at 96.7%; (3) the whitening preprocessor that rescues collapsed spaces (GPT-2: 0.130 → 0.478); (4) the operating rule inherited by any future hub: $\text{hub_width} < \text{dim}(\text{source encoder})$, with ridge or truncation as the lift. **Open, stated:** reverse-direction transfer (ConvNeXt as target), benchmark B1, other languages and domains, non-linear hubs.